Power

File: power.kicad_sch

CM5

File: cm5.kicad_sch

USB_Hub

File: usb.kicad_sch

Audio

File: audio.kicad_sch

Zigbee

File: zigbee.kicad_sch

Ethernet_SD

File: eth_sd.kicad_sch

UI_Mech

File: ui.kicad_sch

HVH-1 Home Voice Hub — Raspberry Pi Compute Module 5 carrier for Home Assistant OS + Claude voice assistant.

Interfaces: USB-C power (5V/5A PD via CM5), 1x USB3.0-A, 1x USB2.0-A, Gigabit Ethernet, microSD boot,

Wi-Fi 5 (2.4/5GHz) + BT5 on CM5, Zigbee/Thread MGM210P, USB audio (MEMS mic + 2.5W speaker amp).

All nets are global labels. Generated from gen/design.py — edit there, then re-run gen/build.sh.

Top level / sheet index

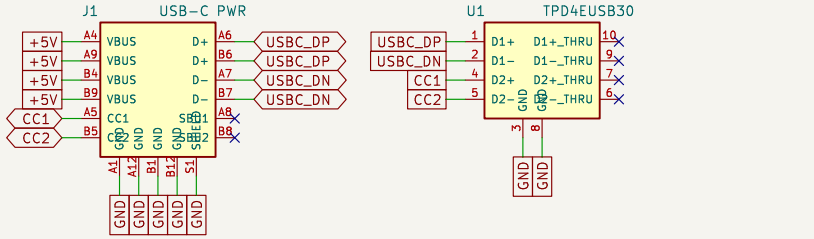
Open hardware — generated design

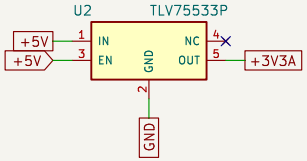
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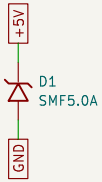
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Title: HVH-1 Home Voice Hub (CM5 carrier)

Size: A3	Date: 2026-11-10	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/8





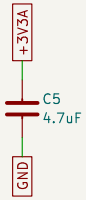


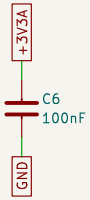
















USB-C: CC1/CC2 go straight to the CM5 PMIC, which negotiates 5V/5A (Raspberry Pi 27W PSU).
Other USB-C chargers give 5V/3A; set usb_max_current_enable=1 (done by the SD-card installer).

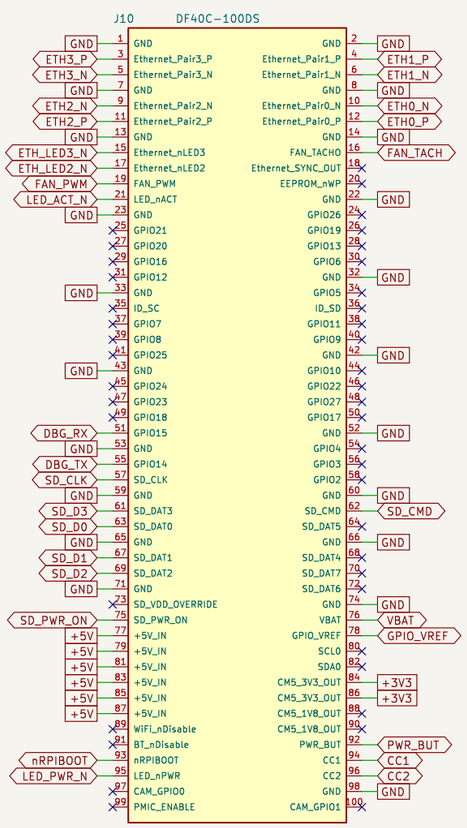
Power input / USB-C / 3.3V

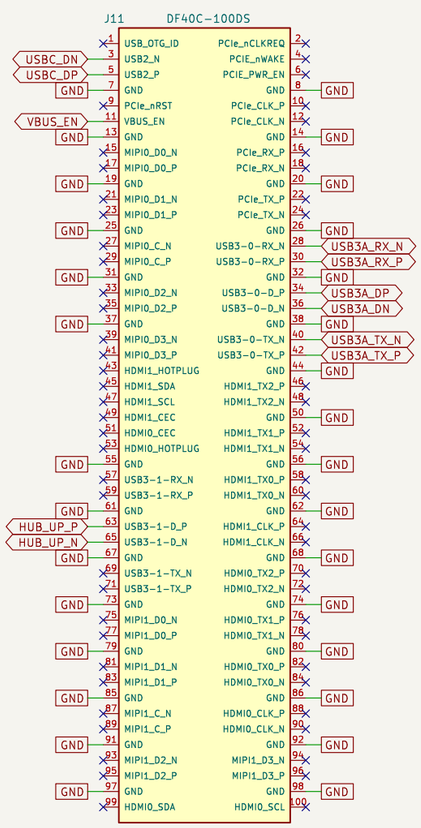
Open hardware — generated design

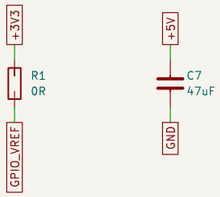
Sheet: /Power/
File: power.kicad_sch

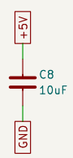
Title: HVH-1 Home Voice Hub (CM5 carrier)

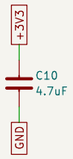
Size: A4	Date: 2026-11-10	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 2/8

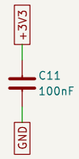


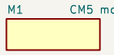


















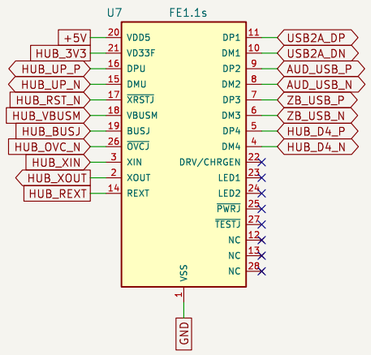
Use CM5 *Lite* (no eMMC) wireless variant, e.g. CM5104000. Boots from J5 microSD.
DF40 pad geometry = HA Yellow / CM4IO. Antenna end of module must stay at board edge.

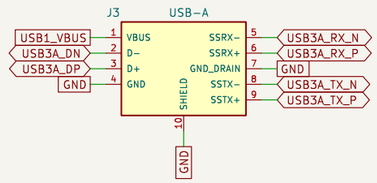
Raspberry Pi CM5 connectors
Open hardware — generated design

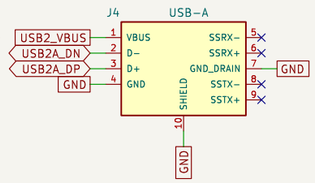
Sheet: /CM5/
File: cm5.kicad_sch

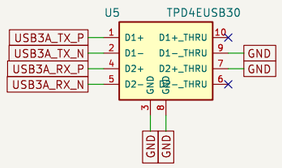
Title: HVH-1 Home Voice Hub (CM5 carrier)

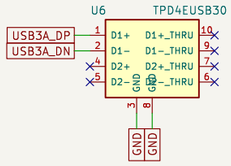
Size: A3	Date: 2026-11-10	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 3/8

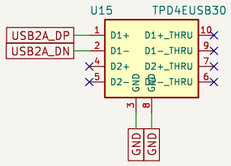


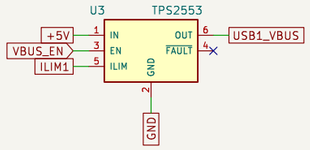


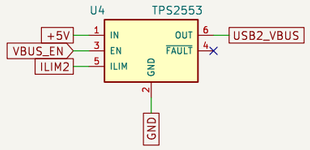


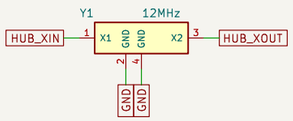


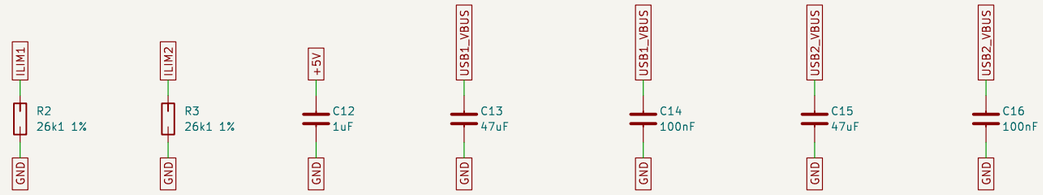


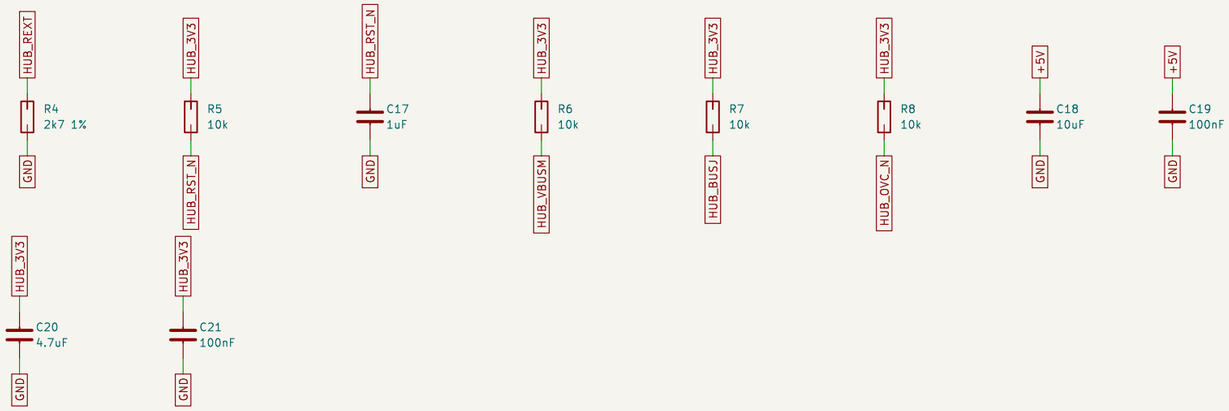


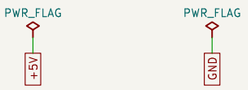












J3 = USB 3.0 (CM5 USB3-0). J4 = USB 2.0 via hub (port 1). Hub port 2 = audio codec, port 3 = Zigbee USB-UART, port 4 = test pads. FE1.1s has internal crystal load caps.

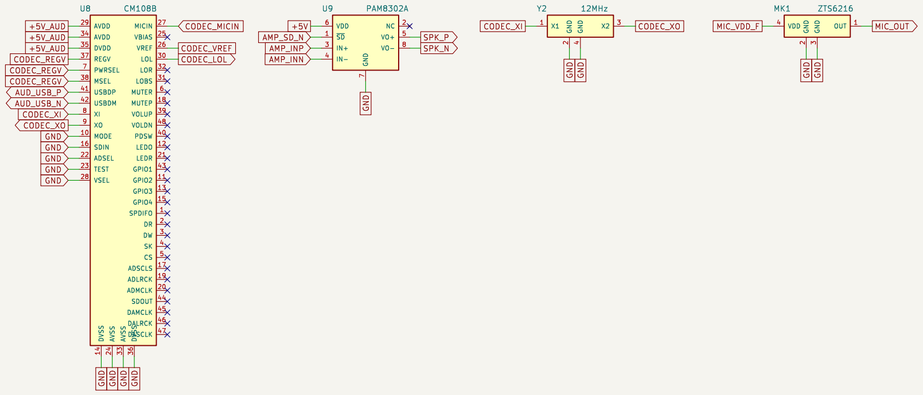
USB-A ports and USB 2.0 hub

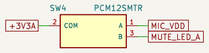
Open hardware — generated design

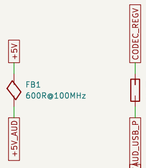
Sheet: /USB_Hub/
File: usb.kicad_sch

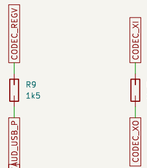
Title: HVH-1 Home Voice Hub (CM5 carrier)

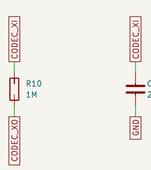
Size: A3	Date: 2026-11-10	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 4/8

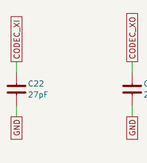


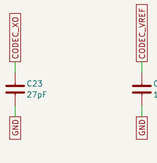


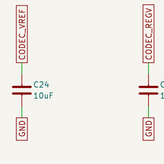


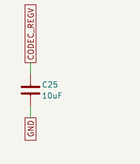


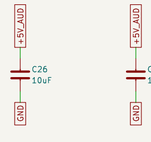


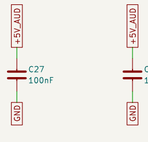


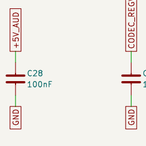


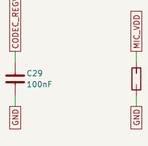


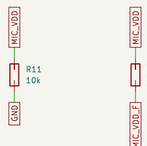


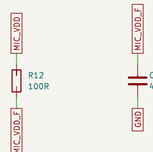


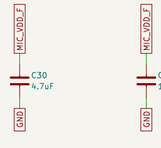


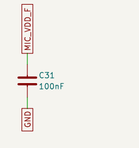


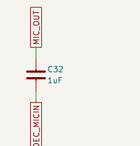


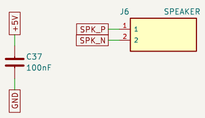














CM108B appears as a standard USB Audio Class device (snd-usb-audio, in HAOS).
SW4 cuts microphone power (hardware privacy) and lights D5 when muted.

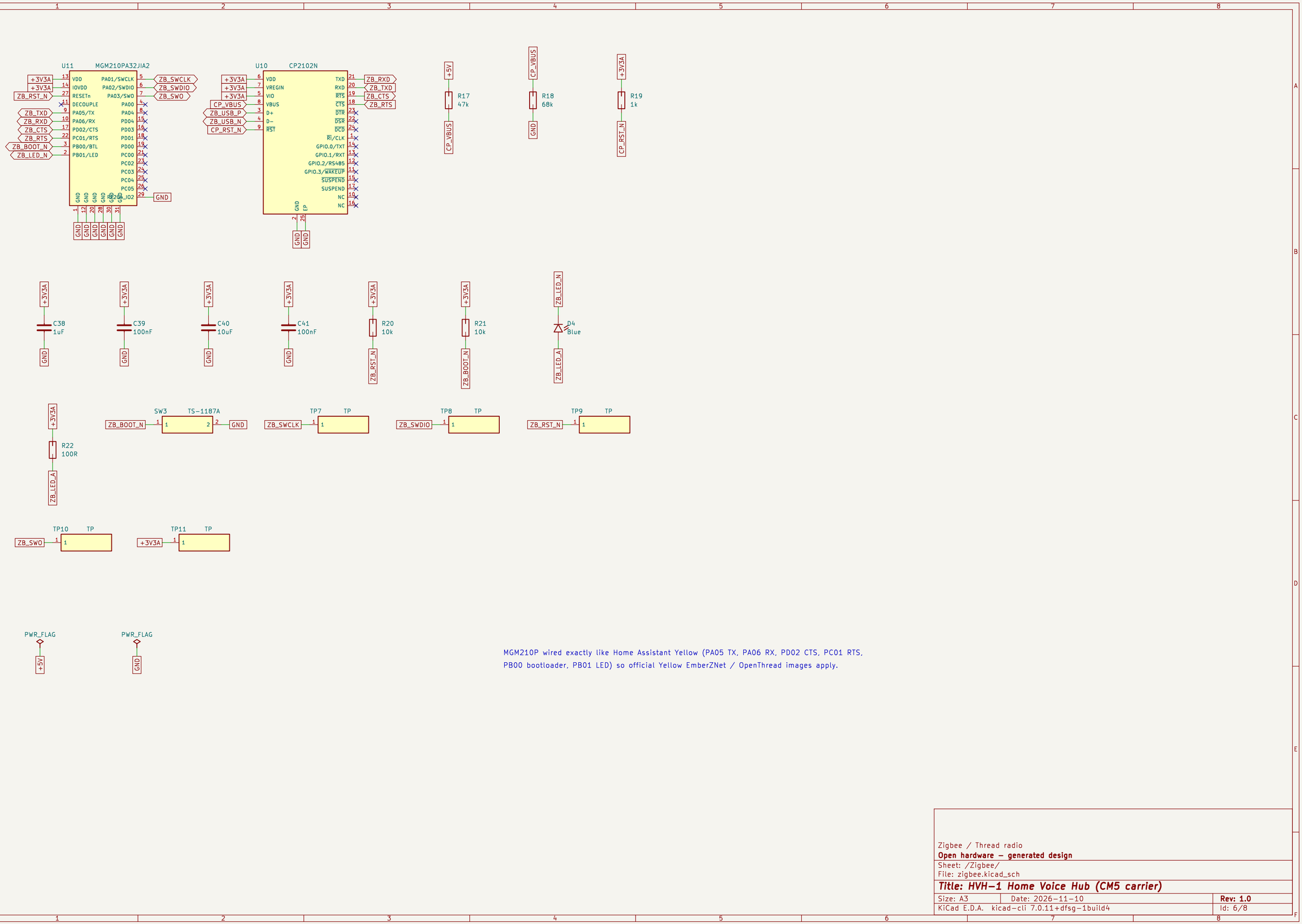
USB audio: microphone and speaker

Open hardware — generated design

Sheet: /Audio/
File: audio.kicad_sch

Title: HVH-1 Home Voice Hub (CM5 carrier)

Size: A2	Date: 2026-11-10	**Rev: 1.0**
KiCad E.D.A.	kicad-cli 7.0.11+dfsg-1build4	Id: 5/8



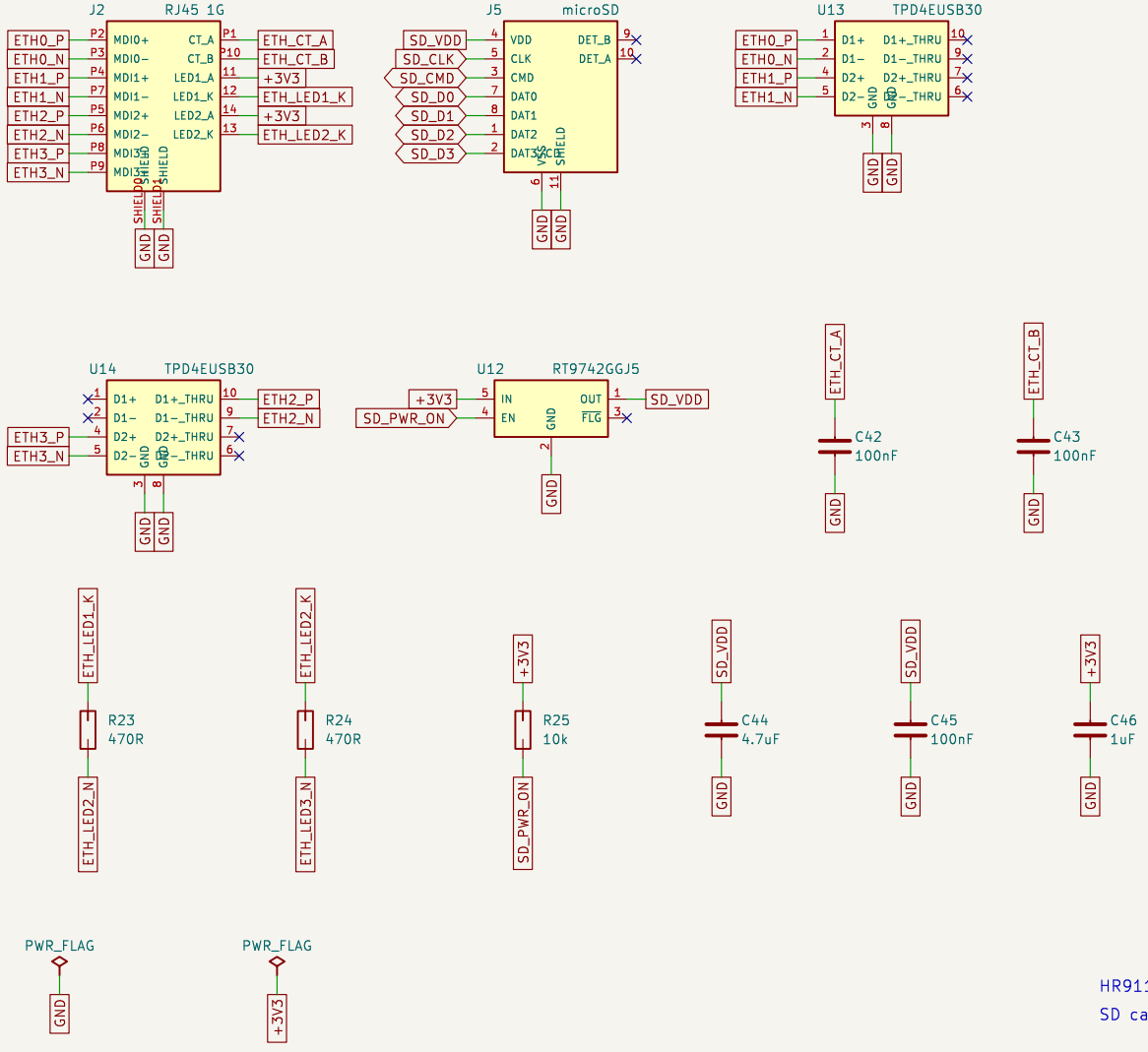
Zigbee / Thread radio

Open hardware — generated design

Sheet: /Zigbee/
File: zigbee.kicad_sch

Title: HVH-1 Home Voice Hub (CM5 carrier)

Size: A3	Date: 2026-11-10	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 6/8



HR911130A LED pin polarity (11/14 anode) per LCSC land pattern — verify on first article.

SD card supply switched by CM5 SD_PWR_ON (required for UHS reset).

Gigabit Ethernet and microSD

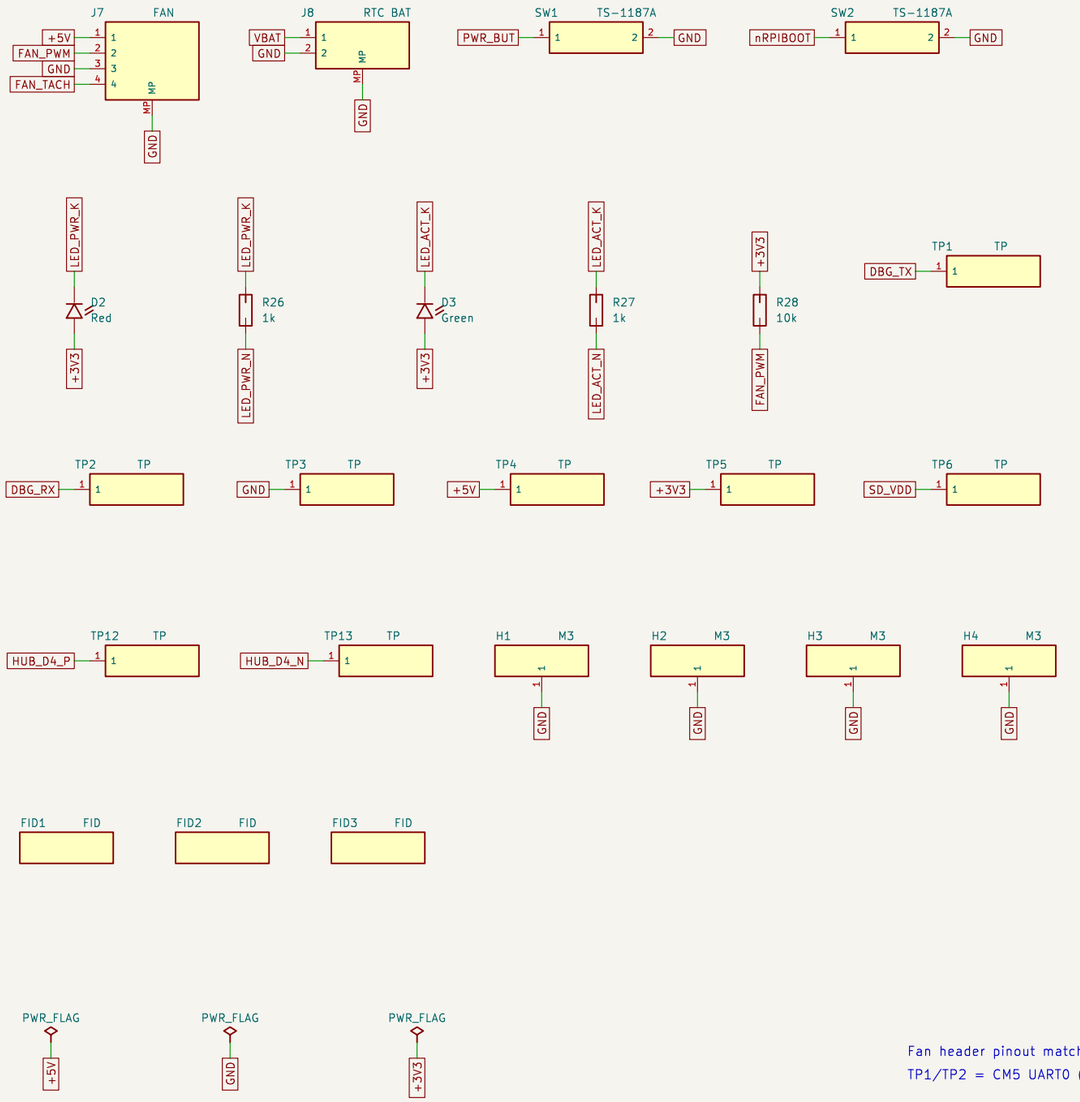
Open hardware — generated design

Sheet: /Ethernet_SD/

File: eth_sd.kicad_sch

Title: HVH-1 Home Voice Hub (CM5 carrier)

Size: A4	Date: 2026-11-10	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 7/8



Fan header pinout matches Raspberry Pi CM5IO (1=5V 2=PWM 3=GND 4=TACH).
TP1/TP2 = CM5 UART0 (GPIO14 TX / GPIO15 RX), 3.3V.

Buttons, LEDs, fan, RTC, test points, mechanics

Open hardware — generated design

Sheet: /UI_Mech/
File: ui.kicad_sch

Title: HVH-1 Home Voice Hub (CM5 carrier)

Size: A3	Date: 2026-11-10	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 8/8